

Q1

- (a) What is natural language processing and why is it important including 3 examples of application?

It is a field of artificial intelligence of how computers hear natural language of written text and spoken language and process it into instructions that a machine can execute. **It is the interaction between computers and human language,** It can be used for machine translation, chat bots dealing with customer queries, virtual assistants like Alexa and sentiment analysis.

There are challenges as computers have difficulty with behaviour like the use of ambiguity, context, dialects, irony, humour and sarcasm and combining this with visual clues from speaker.

- (b) Discuss what is deep learning and why it has become so relevant in recent years?

It is a subset of machine learning that uses **multi-layered neural networks** of input layer over connecting channels to hidden layer and output layer to **process complex data**. It uses **bias** and an **activation** function for processing input . It has become highly relevant with advances in hardware, software algorithm code, datasets and network architecture, enabling applications in many fields.

The recent relevant **of deep learning** is **attributed** to **advances in advances in hardware**, software algorithm code, datasets and network architecture design. This has led to wide spread use in finance, healthcare, self-driving cars and natural language processing.

- (c) Discuss the concept of Entropy and its relation to information gain?

Entropy is a measure of impurity in a dataset.

Information gain, is a metric used in the **decision tree algorithm** to evaluate the effectiveness of splitting the data set on a **particular attribute**. It measures the **reduction in entropy** after a **splitting** of the dataset into different categories based on this attribute.

Q2 (a) Explain the concept of clustering including the two typical examples K-Means and hierarchical clustering.

What is clustering?

Clustering is an **unsupervised learning** technique used to group data points so that:

- Points **within the same cluster are similar** to each other
- Points **in different clusters are dissimilar**

Unlike classification, clustering:

- Has **no labeled outcomes**
- Discovers hidden structure or patterns in data

Typical applications include:

- Customer segmentation
- Document grouping
- Image segmentation
- Anomaly detection

1. K-Means clustering

Concept

K-Means partitions data into **K predefined clusters**. Each cluster is represented by a **centroid** (the mean of points in that cluster).

How it works

1. Choose the number of clusters K
2. Initialize K centroids (usually randomly)
3. Assign each data point to the nearest centroid
4. Recompute centroids as the mean of assigned points
5. Repeat steps 3–4 until assignments no longer change

Key characteristics

- Distance-based (usually Euclidean distance)
- Each point belongs to **exactly one cluster**
- Clusters are roughly **spherical and similar in size**

Advantages

- Simple and fast
- Scales well to large datasets

Limitations

- Must choose K in advance
- Sensitive to initialization and outliers
- Performs poorly with non-spherical or uneven clusters

2. Hierarchical clustering

Concept

Hierarchical clustering builds a **tree-like structure (dendrogram)** that represents nested groupings of data points.

There are two main approaches:

(a) Agglomerative (bottom-up)

1. Start with each data point as its own cluster
2. Iteratively merge the two closest clusters
3. Continue until all points form one cluster

(b) Divisive (top-down)

1. Start with all data points in one cluster
 2. Recursively split clusters
- (Agglomerative is more common.)

Distance between clusters (linkage methods)

- **Single linkage:** minimum distance
- **Complete linkage:** maximum distance
- **Average linkage:** average distance
- **Ward's method:** minimizes within-cluster variance

Advantages

- No need to pre-specify the number of clusters
- Produces an interpretable dendrogram
- Can capture clusters of different shapes and sizes

Limitations

- Computationally expensive for large datasets
- Sensitive to noise and distance metrics
- Once a merge/split is made, it cannot be undone

Comparison: K-Means vs Hierarchical Clustering

Aspect	K-Means	Hierarchical
Type	Partitioning	Tree-based
Number of clusters	Must be specified	Chosen after dendrogram
Scalability	Very good	Poor for large datasets
Cluster shape	Spherical	Flexible
Interpretability	Moderate	High (dendrogram)

Summary

Clustering groups unlabeled data based on similarity. K-Means is efficient and simple for large, well-separated clusters, while hierarchical clustering provides a flexible and interpretable view of data structure but is more computationally expensive.

(b) Discuss the different between filters and wrappers including when it is best to use one over the other.

Filter methods select features based on statistical measures like correlation while **wrapper methods** use model performance as criterion for feature selection Filter method are less computationally demanding and are recommended for high dimensional data. Wrapper methods lead to better performance.

(c) Give the typical steps you will need to create a pipe line of sentiment analysis.

1. Collect data with sentiment labels
2. Preprocess the text by cleaning and transforming
3. Extract numerical features
4. Choose and appropriate model
5. Train and evaluate the model to new data
6. Interpret the results
7. Deploy the model if it performs well

Q3 (a) Discuss the purpose of PCA and when you should use it.

Purpose of Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical technique used to **reduce the dimensionality of data** while preserving as much of the original information (variance) as possible.

It does this by:

Transforming the original, possibly correlated features into a **new set of uncorrelated variables** called *principal components*

Ordering these components so that the first few capture most of the variation in the data

In short, PCA **simplifies data without throwing away too much information**.

When you should use PCA

1. When dealing with high-dimensional data

Use PCA when your dataset has:

Many features

Redundant or overlapping information

PCA helps reduce computational cost and lowers the risk of overfitting.

2. When features are highly correlated

If predictors show strong correlations:

PCA removes multicollinearity

Improves stability of models like linear or logistic regression

3. When you want noise reduction

Lower-variance components often represent noise.

Removing them can:

Improve model generalization

Increase signal-to-noise ratio

4. When model performance matters more than interpretability

PCA is ideal if:

You don't need to explain individual feature effects

You care primarily about prediction accuracy

5. For data visualization

PCA is commonly used to:

Project data into 2D or 3D

Reveal structure, clusters, or outliers

(b) Discuss when the following feature selection methods should be applied: Filter, Wrapper, Deep Learning and Principal Component Analysis.

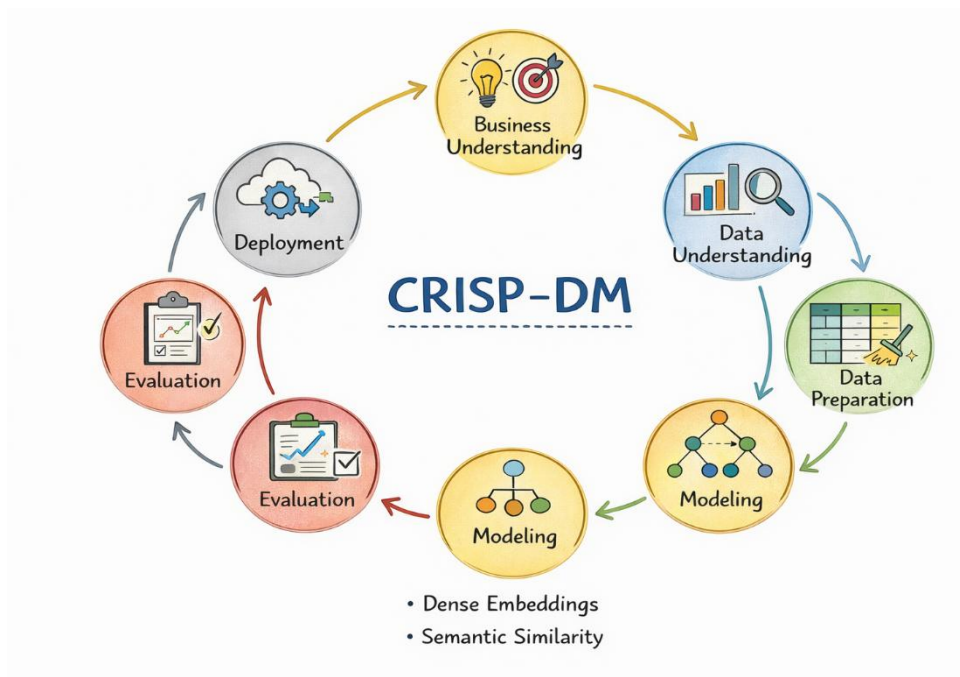
A **filter method** when dealing with large dataset to **quickly eliminate irrelevant** or redundant **features**.

A wrapper method **uses when optimising feature selection for machine learning model, particularly for maximising predictive performance**. Wrapper methods are computationally expensive.

Deep learning: **It is a subset of machine** learning that uses **multi-layered neural networks** of input layer over connecting channels to hidden layer and output layer to **process complex data**. It uses **bias** and an **activation** function for processing input . It has become highly relevant with advances in hardware, software algorithm code, datasets and network architecture, enabling applications in many fields.

PCA: Apply PCA when you **need dimensionality reduction** for **high-dimensional, correlated data** and **predictive performance matters** more than feature interpretability.

(c) Discuss the data preparation step in CRISP-DM process and give examples to two common techniques.



The data preparation step in CRISP-DM involves **checking for missing data**, checking the format like dates or standardising abbreviation, **normalising or scaling data**, **aggregating data and elimination any duplication**. Clean and preprocess the data. Select relevant features and transform the data set.

(d) Discuss concept reinforcement learning and its relationship to Markov Decision Processes.

1. What is Reinforcement Learning?

Reinforcement Learning is a learning paradigm where an **agent** learns to make decisions by **interacting with an environment** in order to **maximize cumulative reward**.

Instead of being told the correct action, the agent:

- Tries actions
- Observes consequences
- Receives rewards or penalties
- Improves its behavior over time

Core idea

Learn by trial and error using feedback from the environment.

2. Key components of Reinforcement Learning

An RL problem is defined by:

Component	Description
Agent	Learner/decision-maker
Environment	Everything the agent interacts with
State (s)	Current situation of the agent
Action (a)	Choices available to the agent
Reward (r)	Scalar feedback signal
Policy (π)	Strategy mapping states $\rightarrow$ actions

3. What is a Markov Decision Process (MDP)?

A **Markov Decision Process** is a **mathematical framework** for modeling sequential decision-making problems where outcomes are partly random and partly under the agent's control.

An MDP is defined as a 5-tuple:

$$(S, A, P, R, \gamma)$$

Element	Meaning
S	Set of states
A	Set of actions
P(s' s, a)	Transition probability
R(s, a)	Reward function
γ	Discount factor

4. The Markov property (crucial link)

The defining assumption of an MDP is the **Markov property**:

$$P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots) = P(s_{t+1} | s_t, a_t)$$

The future depends only on the present state and action, not the past.

This makes planning and learning tractable.

5. Relationship between RL and MDPs

Key relationship

Reinforcement Learning is the solution approach;

Markov Decision Processes are the problem formulation.

- **MDP**: Defines *what* the environment is
 - **RL**: Defines *how* the agent learns to act in it
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Two cases

1. MDP known → Planning

If *P* and *R* are known:

- Use **Dynamic Programming**
- Examples: Value Iteration, Policy Iteration

2. MDP unknown → Reinforcement Learning

If P and/or R are unknown:

- Learn by interaction
 - Examples: Q-learning, SARSA, Policy Gradients
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6. Value functions in MDPs and RL

State-value function

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

Action-value function

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

These quantify **how good** a state or action is.

7. Bellman equations (the bridge)

Bellman equations describe **recursive structure** in MDPs:

Bellman expectation equation

$$V^\pi(s) = \sum_a \pi(a \mid s) \left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s') \right]$$

Bellman optimality equation

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^*(s') \right]$$

RL algorithms approximate these equations **without knowing** P .

8. Example: Gridworld

- States: grid cells
- Actions: up, down, left, right
- Reward: -1 per step, $+10$ at goal
- Transitions: stochastic

This is an **MDP**.

Using RL (e.g., Q-learning), the agent:

- Explores the grid
- Learns Q-values
- Discovers the shortest path to the goal

9. Extensions beyond basic MDPs

Model	When used
POMDP	Partial observability
Semi-MDP	Variable-duration actions
Continuous MDPs	Continuous states/actions
Multi-agent MDPs	Multiple agents

10. Why MDPs matter in RL

MDPs provide:

- Theoretical guarantees
- Convergence proofs
- A unifying language for RL algorithms

Almost all classical RL theory assumes the environment is an **MDP**.

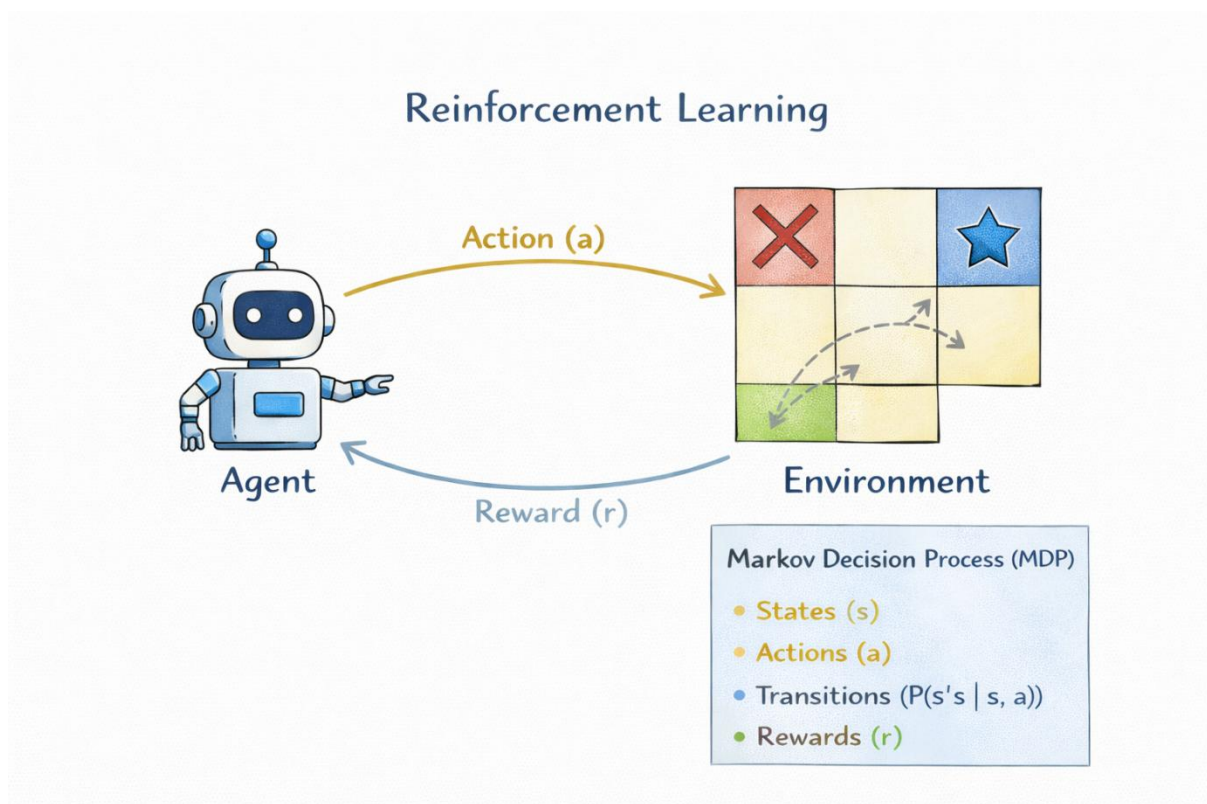
11. Summary (exam-ready)

- **Reinforcement Learning** is a learning paradigm where agents learn optimal behavior via interaction and reward.

- **Markov Decision Processes** provide the mathematical framework for modeling RL environments.
 - RL algorithms aim to solve MDPs **when the model is unknown**.
 - The **Markov property** enables efficient learning and planning.
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One-line takeaway

Reinforcement Learning learns policies; Markov Decision Processes define the world in which those policies act.



(e) Give short definitions related to explainability: accountability Trustworthiness, confidence, causality, fairness ethical decision

Accountability

The ability to **identify who is responsible** for an AI system's decisions, outcomes, and impacts, including mechanisms for oversight and redress.

Trustworthiness

The degree to which an AI system is **reliable, transparent, safe, and aligned with human values**, enabling users to confidently rely on its outputs.

Confidence

A measure of how **certain an AI system is about its predictions or decisions**, often expressed probabilistically (e.g., prediction confidence or uncertainty).

Causality

The extent to which an AI model captures or explains **cause-and-effect relationships**, rather than mere correlations, between inputs and outcomes.

Fairness

The property of an AI system to **avoid unjustified bias or discrimination** and to treat individuals or groups equitably across sensitive attributes.

Ethical Decision-Making

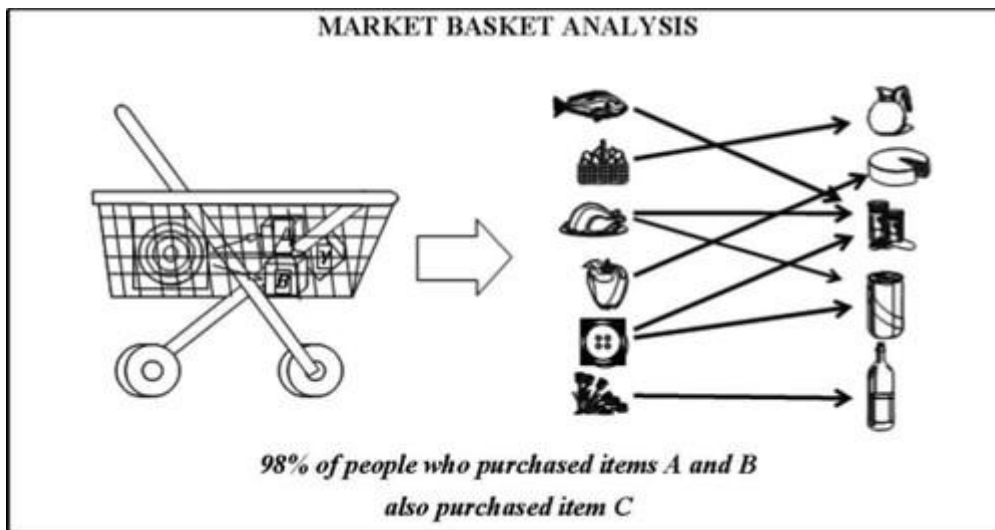
The process by which AI systems (or their designers) make choices that **respect moral principles**, such as beneficence, non-maleficence, autonomy, and justice.

One-line summary

Explainable AI supports **accountability**, builds **trustworthiness**, communicates **confidence**, reveals **causality**, ensures **fairness**, and enables **ethical decisions**.

1. Define association rules and their applications.

Association Rules



$$\begin{aligned}
 \text{Rule } X \Rightarrow Y & \begin{cases} \text{Support} = \frac{\text{Frequency}(X,Y)}{N} \\ \text{Confidence} = \frac{\text{Frequency}(X,Y)}{\text{Frequency}(X)} \\ \text{Lift} = \frac{\text{Support}}{\text{Support}(X) * \text{Support}(Y)} \end{cases}
 \end{aligned}$$

Definition

Association rule mining is a data mining technique used to discover **interesting relationships, correlations, or patterns** among items in large datasets.

An **association rule** is an implication of the form:

$$X \rightarrow Y$$